Mehdi Karami

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Summary

- **Machine Learning Researcher** specializing in **image processing**, proposed a **pre-trained GAN for limited-label brain MRI segmentation** to advance **medical AI research**.
- Strong foundation in **machine learning theory and practice**, with experience in **model development, training, and optimization**, and a clear focus on contributing to **cutting-edge AI projects** and **real-world applications**.
- **Active open-source contributor** in AI, collaborating on **public and volunteer projects**, reflecting commitment to **innovation, collaboration, and advancement of the AI community**.

Education

2021–2025 **Bachelor of Computer Engineering**
Zand Institute of Higher Education
Shiraz, Persia
GPA: 17.88/20 (equivalent to 3.57/4.00)
Research: Medical Image Processing, Brain Image Segmentation

2023–2024 **Visiting Student (Bachelor of Computer Engineering)**
University of Science and Culture
Tehran, Persia
Bachelor's Project: AI Sight Quest (Integration of BERT-SQuAD for on-device document question answering)

Research Interests

- Machine Learning
- Computer Vision
- Deep Learning
- Computational Biology
- Natural Language Processing
- Scientific Computing

Publication

Pre-Trained Generative Adversarial Network for Limited-Labeled Brain MRI Segmentation engrXiv, DOI: 10.31224/6504
Mehdi Karami✉, Betsabeh Tanoori

Open Source Projects

[1] **Retrieval-Augmented Generation (RAG) for Small Language Models (SLMs)**
A RAG pipeline, including text chunking, embedding, and vector retrieval, which enables efficient generation and inference with local Small Language Models (SLMs) as well as standard language models.
Stack: Pytorch, FAISS, Transformers (HuggingFace), Sentence-Transformers, LangChain, PyMuPDF4LLM, ...

[2] **Cyberattack Detection and Anomalous Behavior Analysis**
An LSTM-based RNN for detecting cyberattacks in anomalous network traffic, including DDoS, port scanning, and brute-force attacks.
Stack: TensorFlow/Keras, Scikit-learn, PCA, ROC Curve, ...

[3] **Solar Power Generation Prediction**
Implementing ML models (Linear, Decision Tree, Random Forest, Gradient Boosting, MLP, DNN) to predict solar power generation and evaluate the results via MAE, MSE, R^2, and confusion matrix.
Stack: TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn, ...

[4] **NexumAI**
An interface for training models (e.g., image classification, segmentation) without technical expertise, letting users obtain trained models by supplying datasets and configurations.
Stack: TensorFlow, PyTorch, Torchvision, PyQt6, ...

More projects and contributions: GitHub

Work Experience

2023–2024 **iOS Developer** (Full-time, on-site)
Round Table Apps (Sydney, Australia)
Development Office: Tehran, Persia

Projects:
Blossom – Save and Invest , a financial platform serving **25K+** active users across **Australia** and **New Zealand**.
MailPlus – MP Operator , an **Australian** post-transport application serving **300+** owner-operators and **30K+** customers.
Resilience Box , an **international** wellbeing and resilience application.

Technical Contributions:
Analyzed and implemented **data flows** to model **user behavioral patterns** across various scenarios.
Applied key design patterns such as **MVC**, **MVVM**, **VIP**, and **Repository**.
Refactored legacy applications to improve maintainability, streamline **API integration**, and support future updates.
Research and development of **AR** and **VR** technologies in **healthcare** using **visionOS**.
Boosted development efficiency by up to **40%** through reusable templates for modules and repository implementations.

Teaching Experience

2024–2025 **Teaching Assistant, Zand Institute of Higher Education**

Winter 2025 Fundamentals of Computer and Programming
Winter 2025 Digital Electronics
Winter 2025 Computer Architecture
Fall 2024 Human-Computer Interaction
Fall 2024 Digital Electronics

Technical Skills

Programming & Scripting: Python, C/C++, Julia, MATLAB, Swift, Objective-C, SQL, Bash, LaTeX, ...

Data Science & ML: TensorFlow/Keras, PyTorch, Scikit-learn, OpenCV, HuggingFace, NumPy, Pandas, ...

Deep Learning: Neural Networks, Image Processing, Natural Language Processing, Transformers, GAN, RAG

Software Development: Git, CI/CD, Design Patterns, SOLID, iOS Development, Unit Testing, TDD, BDD, Agile

Tools & Platforms: Jupyter, Google Colab, GitHub, GitLab, Command Line, Linux, macOS, Jira, Confluence, Trello

Databases & Cloud: MLOps, Vector DBs, Relational DBs (SQLite, PostgreSQL), Docker, Kubernetes, Query Optimization

Volunteering

Dec 2024 **Workshop on Git and GitHub, Zand Institute of Higher Education**, Certificate

May 2024 **Workshop on Git and GitHub, K. N. Toosi University of Technology**, Certificate

Honors & Awards

May 2022 **First Place Winner in the C++ Programming Language Competition, Zand Institute of Higher Education**

Hobbies

• Swimming • Bodybuilding • Traveling • Gaming